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Maternal Immune Activation and Schizophrenia:

Investigating Alterations in Social Interaction

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Abstract

Bacterial infection results in activation of the immune system. Maternal immune activation (MIA) during pregnancy, such as administering lipopolysaccharide (LPS) of *E. coli*, a bacterial endotoxin, into pregnant Sprague-Dawley rats can alter the development of the foetal brain, increasing the risk of schizophrenia-like behaviour in adulthood. Previous work suggests that there are specific periods of gestation that increase the risk of developing deficits; specifically, deficits in social interaction. In the current study, LPS was administered in early, mid or late gestation to examine whether the timing of immune activation was associated with schizophrenia-like social interaction deficits. Although the offspring exposed to LPS in mid-gestation produced social withdrawal behaviours; this effect was only evident in adolescent rats. In contrast, we revealed offspring exposed to LPS in late gestation had increased locomotor activity in adulthood. The findings show that the timing of infection affects the development of social deficits, suggesting that infection during mid-gestation is a vulnerable period that may lead to future development of schizophrenia.

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Maternal Immune Activation and Schizophrenia:

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Schizophrenia is a debilitating psychiatric disorder, characterised by positive, negative and cognitive symptoms (Patterson, 2009). Global prevalence is around 1% (Tandon, Keshavan & Nasrallah, 2008); and this disorder can have profound effects on an individual's ability to function well in society. Onset typically occurs during late adolescence and strongly reduces quality of life by decreasing both social and employment prospects (Marwaha et al., 2007; Mendrek & Mancini-Marie, 2015; Perkins & Rinaldi, 2002) as well as increasing the risk of substance use disorder (McCreadie, 2002), suicide (Bushe, Taylor & Haukka, 2010) and other co-morbid psychological disorders (Buckley, Miller, Lehrer & Castle, 2009). Furthermore, schizophrenia diagnoses contribute to family stress (Bland & Foster, 2012), and strain on criminal justice, social welfare and health care systems, from repeated hospitalisations, prescribed medications and increases use of outpatient services. As a result, the public, emotional and financial cost of schizophrenia impacts individual families and society (Montgomery et al., 2013; Olesen, Gustavsson, Svensson, Wittchen & Jönsson, 2012).

Despite the cost to both individuals and society, both the aetiology and most effective treatment protocols are not well understood. In fact, even the diagnostic criteria for schizophrenia can differ wildly between individuals. Common symptoms of schizophrenia are divided into three subcategories; *positive symptoms* e.g. delusions, hallucinations or disorganised speech; *negative symptoms* e.g. avolition (lack of motivation), alogia (lack of verbal communication), anhedonia (lack of pleasure from enjoyable activities), blunted affect (reduced emotive expression) and asociality (social withdrawal); and *cognitive symptoms* e.g. deficits in attention,

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abstract thinking, learning, memory or cognitive flexibility (American Psychiatric Association [APA], 2013). To achieve a diagnosis of schizophrenia, two or more of the following symptoms must be present for an extended period of time; delusions, hallucinations, disorganised speech, disorganised or catatonic behaviour and any of the negative symptoms. Thus, one individual may just have all three positive symptoms present, while another may present with two positive symptoms along with multiple negative and cognitive symptoms, resulting in a wide variety of possible symptom presentations between patients. This heterogeneous nature of schizophrenia results in a variety of symptom combinations, onset patterns and differences in the severity of the disorder and responses to treatment which in turn makes it difficult to identify valid neurodevelopmental theories and effective treatments.

Genome-wide association studies have identified over 100 specific genetic variants associated with an increased risk of developing schizophrenia (Ripke et al., 2014). Studies using twins have shown that despite monozygotic twins sharing 100% of their genetic code, only 65-85% of the variance is accounted for by genetic factors, hence, it is possible for only one twin to develop schizophrenia in later life. This suggests that other factors such as environmental influences contribute to the manifestation of symptoms (Lichtenstein et al., 2009; Sullivan, Kendler & Neale, 2003; Wray & Gottesman, 2012).

In line with this, migration, urban birth, low socioeconomic class, maternal famine, obstetric complications, foetal hypoxia and maternal exposure to infection, have all been shown to increase the risk of developing schizophrenia in the presence of a genetic predisposition (Jablensky, Morgan, Zubrick, Bower & Yellachich, 2005; Khandaker, Zimbron, Lewis & Jones, 2013; Sorensen et al., 2014; Walder et al., 2014). However, many studies fail to address the interaction between these genetic or

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environment influences (Khandaker et al., 2013). Even during gestation, monozygotic twins can have separate intrauterine environments (shared or individual placenta and amniotic sac) and childbirth experience can differ between children (Jablensky et al., 2005; Sela & Flood, 2012). Overall, this suggests that while a genetic predisposition may *enhance* the risk of developing schizophrenia, it does not explain the entire aetiology (Tandon, Keshavan & Nasrallah, 2008).

One possible explanation, the *neurodevelopmental hypothesis*, suggests that disruption in early brain development, *in utero* or in the perinatal period, can result in a vulnerability to poor psychological functioning in adulthood (McGrath, Féron, Burne, Mackay-Sim & Eyles, 2003). Interest in prenatal environmental challenges was initiated by Stein, Susser, Saenger and Marolla (1975), who reported increased behavioural and health conditions in those born during, and immediately after the Dutch Hunger Winter (1944-1945). The severity, duration and regions affected by the famine were measurable, which allowed comparison with controls not exposed to malnutrition, infection and stress during this period. Hoek, Brown and Susser (1998) later reported that those exposed to famine during the Hunger Winter had an increased risk of developing schizophrenia.

Similarly, Bradbury and Miller (1985) reported a season-of-birth effect, whereby individuals with schizophrenia are more likely to be born late winter to early spring and therefore *in utero* during winter. The prevalence of schizophrenia also increased following the Asian 1957 influenza pandemic (Mednick, Machon, Huttunen & Bonett, 1988). Clinical and animal research has since accumulated indicating that the risk of infection is high during these periods, suggesting maternal infection during gestation is contributing to an increased risk of developing schizophrenia in offspring.

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Specific periods have been identified when an individual is particularly vulnerable to environmental risk factors. First, maternal infection and maternal stress are well-established risks during the prenatal period, which begins at conception and lasts throughout gestation (Khandaker et al., 2013). Second, obstetric complications and health of child at birth are risks associated with the neonatal period, spanning the first 28 days of life (Apgar, 1966; Kotlicka-Antczak, Gmitrowicz, Sobów & Rabe-Jabłońska, 2001). Lastly, adolescence encompasses children aged 10 to 19 and is another vulnerable phase of brain development; during this period the risk of schizophrenia is exacerbated by exposure to psychoactive substances and emotional stress (Brady & Sinha, 2005). In summary, a wide variety of genetic and environmental interactions result in the development of schizophrenia. Therefore, identifying at risk individuals remains a complex task.

These environmental risks are difficult to empirically study due to family reports of early environment being often subjective, retrospective and often inaccurate (Brown & Lumley, 1998). In addition, birth cohort designs provide no evidence which specific mothers were exposed to infection and whether it was their children who later developed schizophrenia. Therefore, prospective cohort studies aim to minimise these limitations by recruiting pregnant females and medical professionals to collect data as it arises (Brown et al., 2000). Demographic details and maternal serum are collected with maternal and foetal health records throughout pregnancy, childbirth and development. Using such methods, prenatal maternal infection during first and second trimester or respiratory infection during second trimester increases the risk of the offspring developing schizophrenia (Brown et al., 2000; Brown et al., 2004; Cannon et al., 2000).

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While clinical research provides specific insight into the affected population, animal research with rats enables specific risk factors to be manipulated in a controlled environment, such as the introduction of infection at specific doses or times during gestation. Polyriboinosinic - polyribocytidilic acid (PolyI:C) or lipopolysaccharide (LPS) is predominantly used to simulate viral or bacterial infections in animal models, respectively (Fortier, Luheshi & Boksa, 2007; Patterson, 2009). Administration of LPS of *Escherichia coli* (E.coli), a bacterial endotoxin, activates the immune system in exposed animals to produce increased quantities of pro-inflammatory cytokines, such as interleukin-8 (IL-8) and interleukin-6 (IL-6), designed to fight the LPS-induced bacterial infection (Gayle et al., 2004; Smith, Li, Garbett, Mirnics & Patterson, 2007). This response is therefore helpful for the protection of the mother and the foetus from the infection agent; however, this can, at the same time, cause damaging levels of inflammation during this vulnerable period of brain development. As such, Zuckerman, Rehavi, Nachman and Weiner (2003) hypothesised that risk of schizophrenia increases due to maternal immune activation (MIA). Bilbo and Schwarz (2009) suggest this is due to a priming effect on the foetus, resulting in immune system vulnerability throughout life. Hence, the relationship between MIA and schizophrenia is well established in clinical and animal research; however, the relationship between time of infectious exposure and resulting behavioural deficits has not yet been clearly identified (Meyer, Yee & Feldon, 2007).

Exposure to LPS during early gestation has resulted in schizophrenia-like behaviour, such as deficits in prepulse inhibition, latent inhibition and working memory (GD10/11, Waterhouse, Roper, Brennan & Ellenbroek, 2016) and spatial learning and memory (Hao, Hao, Li & Li, 2010). Interestingly, Hao, Hao, Li and Li found that deficits increased as age increased, models the prodromal presentation of

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symptoms and a worsening of symptom severity over time. Deficits in social behaviour have also been reported in offspring following one exposure to LPS at gestational day (GD) 9.5 (Hava et al., 2006; Kirsten, Taricano, Maiorka, Palermo-Neto & Bernardi, 2010).

In addition to the effects of early MIA, LPS-exposure during mid-gestation also results in schizophrenia-like traits. In fact, one LPS injection on GD 15 produced play behaviour deficits in adolescent rats (Taylor, Veenema, Paul, Bredewold, Isaacs & De Vries, 2012). One injection on two consecutive days (GD15/16) produces a wide-range of social and cognitive deficits in adolescent offspring, such as reduced acoustic startle response, exploration behaviours, recognition memory, locomotor activity (Oskvig et al., 2012; Lowe et al., 2008). However, Wischhof, Irrsack, Osorio and Kock (2014) found conflicting results that LPS-exposure on GD15/16 results in locomotor hyperactivity. This finding is particularly interesting due to locomotor changes being present in adolescence (PND33) and adulthood (PND60).

The neurochemical effects of LPS-exposure during late gestation (GD18/19) are well-established in the literature; however, little research investigates behaviours associated with this MIA. Fortier, Luhshi & Boksa (2007) report LPS injections on GD18/19 produced deficits in acoustic startle response; however, it is unclear whether LPS-exposure on GD18/19 alters social behaviours in the offspring.

Overall, the current literature demonstrates that prenatal LPS exposure does result in behavioural deficits; however it remains unclear what effect LPS has on active social interaction and social preference for novelty. In addition, it remains unclear whether there are specific vulnerable stages in prenatal development, whereby the timing of LPS-exposure determines the type or severity of behavioural deficit (Bilbo & Schwarz, 2009; Oskvig, Elkahloun, Johnson, Phillips & Herkenham, 2012).

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Currently, the most frequently used paradigms in the literature focus on attention, working memory and executive functioning; however, less emphasis is placed on the effect of MIA on social cognitive behaviours (Harvey & Boksa, 2012). Yet, social behaviours are very important traits to investigate and are a core part of the negative symptoms of schizophrenia. In the human population, presentation of negative symptoms predicts poor prognosis, decreases quality of life and reduces working and socialising opportunities (Bengtsson-Tops & Hansson, 2001; Rüesch, Graf, Meyer, Rössler & Hell, 2004). The importance of these symptoms is further emphasised by the fact that current treatment options are virtually ineffective in improving such symptoms (Leucht, Corves, Arbter, Engel, Li & Davis, 2009; Lieberman et al., 2005).

The aim of the present study was to investigate if exposure to prenatal bacterial infection during early, middle or late gestation produces deficits in social interaction. To test this, bacterial infection was induced in the pregnant female (dam) using a single subcutaneous LPS injection (0.5 mg/kg) on two consecutive days, either in early (GD 10 and 11), mid-term (GD 15 and 16) or late gestation (GD 19 and 20). Control dams were injected with Saline on the same gestational days. This model was used due to prior success in producing schizophrenia-like deficits in LPS-exposed offspring (Fortier et al., 2007; Waterhouse et al., 2016). Specifically, Waterhouse et al. (2016) found cognitive deficits after LPS exposure on GD10/11 but not GD15/16 or GD18/19.

To assess the effect of LPS-exposure during development, offspring social behaviours will be tested during adolescence (PND21 - 30) and adulthood (PND 60 – 70) (Bilbo & Schwarz, 2009; Hao et al., 2010; Wischhof et al., 2014).

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Offspring will be exposed to two social behaviour paradigms, *play behaviour* and *social approach avoidance*, to capture changes in active social interaction and preference for social novelty, respectively. It is predicted that if MIA in early gestation produces social interaction deficits, then offspring exposed to LPS on GD10/11 and GD15/16 will show reduced active play behaviour. This behavioural change is expected in adolescent and adult offspring. Secondly, it is predicted that if MIA in early gestation does produce social interaction deficits, then offspring exposed to LPS on GD10/11 and GD15/16 will show avoidance behaviours from novel social stimuli. This behaviour change is expected in adolescent and adult offspring. Such findings would provide support for the hypothesis that prenatal MIA interferes with foetal neurodevelopment resulting in schizophrenia-like social cognitive deficits.

Method

Subjects

Twenty-four (18 female and 6 male) Sprague-Dawley rats were randomly selected as breeding animals from the existing colony in the Victoria University vivarium. Breeding animals were housed in same-sex groups of no more than three and were briefly handled each week when their cages were changed and during pairing but were otherwise naïve. The vivarium was humidity (~55%) and temperature controlled ($21^{\circ}\text{C} \pm 1$) and maintained on a 12-hour reverse night/day cycle (lights on at 1900). Standard lab chow and water was available *ad libitum*. The Victoria University of Wellington Animal Ethics Committee granted approval for all methods and procedures used in this study.

Each breeding cycle, male-female breeder pairs were housed in a standard polycarbonate cage on metal grids. The bases of cages were examined daily for the

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presence of vaginal mucus plugs which indicates mating behaviour had occurred. When found, this was considered gestational day zero (Oskvig et al., 2012). On average this took 2.3 days ($SD = 1.7$). Pregnant females (dams) were then removed from the grid and individually housed until the birth of their litter. Offspring remained with the dam until post-natal day (PND) 21, then weaned into sex-type groups. Initially offspring were housed in groups of up to six until they reached 200-250 grams and then in groups of three or four thereafter for the duration of the testing.

If a mucus plug was not found after 14 days, breeding pairs were returned to breeding stock for one more cycle before being removed from breeding stock. Successful female breeders had a stand down period of four weeks after the birth of their litter before being recycled back to breeding stock.

Prenatal Treatment. Pregnant dams were randomly assigned to one of six conditions. All dams were injected with a single subcutaneous injection on two consecutive days; three groups of dams received *e. coli* lipopolysaccharide (LPS; 0.5 mg/kg) and three groups received saline. Injections either occurred on gestational days 10 and 11 (5 saline, 3 LPS), gestational days 15 and 16 (4 saline, 7 LPS) or gestational days 18 and 19 (3 saline, 4 LPS).

Experimental Animals. Male offspring of treated dams were used as experimental animals ($n = 125$). The experimental animals did not undergo any direct LPS or saline treatment and therefore will be referred to using their mother's prenatal treatment condition. A randomised selection of male offspring were utilised for data collection at each time point. All experimental animals were injected with a radio-frequency identification transponder (ID-100US, Kent Scientific Corporation, US) for identification purposes between PND 40 and 70.

Procedure

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Testing always occurred in the same testing room between 0800 and 1430 hours to ensure consistent activity levels within groups. Animals were habituated in a clean polycarbonate cage to the experimental room for 15 minutes prior to testing. All test periods were visually recorded using a ceiling-mounted camera and analysed using EthoVision XT (Noldus, The Netherlands).

Social Interaction. Social interaction was assessed twice, first between PND 20 and 25 and then again between PND 60 and 65. The same procedure was followed during both assessments.

A testing arena (40 x 25 x 30cm) was cleaned with 70% ethanol and paper towels, then a thin layer of bedding was spread evenly throughout. Habituation and testing phases were conducted in dark conditions; a single red light suspended 60 centimetres above the play arena facilitated video recording.

After being habituated to the room, two animals from the same litter were habituated together in the play arena for 10 minutes. The animals were then placed in separate cages in the experimental room for 30 minutes of social deprivation before being placed back in the play arena together to complete a 10 minute test phase.

Each time animals were removed from the play arena, excrement was removed and new bedding added when required. The play arena was cleaned with ethanol and paper towels and new bedding was provided for the testing of each new litter.

To ensure consistent assessment of animal behaviour, one observer manually scored active and passive social behaviours displayed during the test phase. Active behaviour was defined as any behaviour where at least one animal actively engages in social interaction and included, but was not limited to, sniffing, grooming, following, mounting, kicking, boxing, wrestling and crawling under or over the partner. Passive

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behaviour included sitting or lying with bodies in contact (File & Hyde, 1978), but without any active social investigation by either animal.

Social Approach and Avoidance. Social approach and avoidance was assessed using a black T-maze twice, firstly between PND 25 and 30 and again between PND 65 and 70. The same procedure was followed during both assessments.

The T-maze (160cm x 90cm x 45cm) was cleaned with 70% ethanol and paper towels. Standard room lighting was used during habituation and testing phases to increase visibility for video recording.

Male untreated Sprague-Dawley juveniles (PND 15-20) from the breeding stock at the Victoria University vivarium were used as social stimuli and were habituated to the experimental room for 15 minutes prior to testing.

Two wire mesh cylinders (11cm diameter x 14cm) were placed 115 cm apart, one at the end of each arms of the T-Maze and weighed down by a ceramic weight to prevent the caged juvenile or the experimental animal from moving the cylinder. In phase one, the experimental animal is placed at the trunk of the T-Maze to freely explore the arena for 10 minutes. Then, the experimental rat is removed and excrement and urine are wiped away using paper towels. In phase two, the first juvenile is placed under a wire mesh cylinder. The experimental rat is then placed back into the base of the T-Maze to freely explore for a further 10 minutes. Again, the experimental rat is removed and excrement and urine are wiped away using paper towels. In phase three, the first juvenile, now familiar to the experimental animal, remains in the same cylinder and the second novel juvenile is placed under the remaining wire mesh cylinder. The experimental animal is placed at the base of the T-Maze for the 10-minute test period.

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The positioning of familiar stimulus was counterbalanced across trials, that is, the familiar stimulus could be placed in the left or right arm of the T-Maze.

Movement of the experimental animal and the duration of time spent in the familiar or novel arm was recorded using a ceiling mounted camera and analysed by Ethovision XT (Noldus, The Netherlands). The frequency of arm entrances and the latency to first entrance was also recorded. Other behaviours, such as sniffing the familiar and the novel stimulus, grooming and the preference of first entry were manually coded by one observer, again, ensuring consistency of assessment across all trials.

Drugs

Lipopolysaccharide of e. coli bacteria (escherichia coli 0111:B4; Sigma-Aldrich, UK) was dissolved in 0.9% sterilised saline and injected at 0.5mg/kg.

Statistical Analyses

All statistical analyses were conducted using SPSS (IBM v22). Our alpha level was set to .05, such that, p-values less than or equal to this indicated significance.

Due to non-normal data, our response variables for play behaviour and social approach and avoidance were all subject to a natural log transformation. All analyses were run with these transformed variables.

One-way analyses of variance (ANOVA) were used to analyse data. The between subjects factors were prenatal treatment (LPS vs saline) and gestational day of injection (GD10/11 vs GD15/16 vs GD18/19). The dependent variables in the play behaviour paradigm were duration of active interaction, duration of passive interaction, frequency of active interaction, frequency of passive interaction, latency to first active interaction and latency to first passive interaction. The dependent variables in the social approach and avoidance paradigm were categorised into three measurements. Firstly, the duration of time spent in the novel arm, in the familiar arm,

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sniffing novel stimulus, sniffing familiar stimulus and duration in the non-social area. Secondly, the frequency of times entering into the novel arm and the familiar arm and the frequency of times sniffing the novel stimulus and the familiar stimulus. Lastly, the latency to first entry into the novel arm and the familiar arm and the latency to the first sniff of the novel stimulus and first sniff of the familiar stimulus. Distance travelled was also recorded.

Repeated measures analysis could not be run to compare adolescent and adult behaviours. Rat's treatment conditions were ascertainable due to housing conditions, but their individual identifications were not. As we were unable to identify data retrieved from each specific animal, we could not run repeated-measures analysis of time-point (adolescence vs. adulthood). As such, comparison was only carried out within each time point, and not across. Post hoc comparisons using the Tukey HSD test were used to identify which groups showed differences in behaviour.

Results

Preliminary analyses indicated minimal, non-significant behavioural differences between saline GD10/11, GD15/16 and GD18/19 conditions (data not shown); therefore, the three conditions were collapsed to increase sample size of the control group (Table 1).

Table 1.

Prenatal Treatment Condition Sample Sizes

Test	LPS GD10/11	LPS GD15/16	LPS GD18/19	Saline
Play Behaviour	7	8	2	15
Time point one				
Social Approach	11	12	3	46

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Avoidance				
Time point one				
Play Behaviour	8	13	3	13
Time point two				
Social Approach	7	7	6	20
Avoidance				
Time point two				

Play Behaviour

Time point 1. Figures 1, 2 and 3 show the time spent engaged in active or passive play behaviour, the frequency of these events and the latency to engage in these behaviours in the play behaviour paradigm. One-way ANOVAs revealed no significant differences at time point one between conditions in active duration ($F(2, 28) = 2.26, p = .103$), passive duration ($F(2, 28) = 1.51, p = .232$), frequency of passive interaction ($F(2, 28) = .158, p = .924$), latency to first active interaction ($F(2, 28) = .572, p = .638$) and latency to first passive interaction ($F(2, 28) = 2.86, p = .055$). However, one-way ANOVA did reveal a significant difference between conditions in the frequency of active behaviour ($F(2, 28) = 4.39, p = .012$); specifically, LPS GD18/19 ($M = 27.50, SD = 9.20$) showed significantly more frequent active behaviour than LPS GD10/11 ($M = 8.71, SD = 6.34$), LPS GD15/16 ($M = 7.50, SD = 5.43$) and saline conditions ($M = 11.00, SD = 8.04$).

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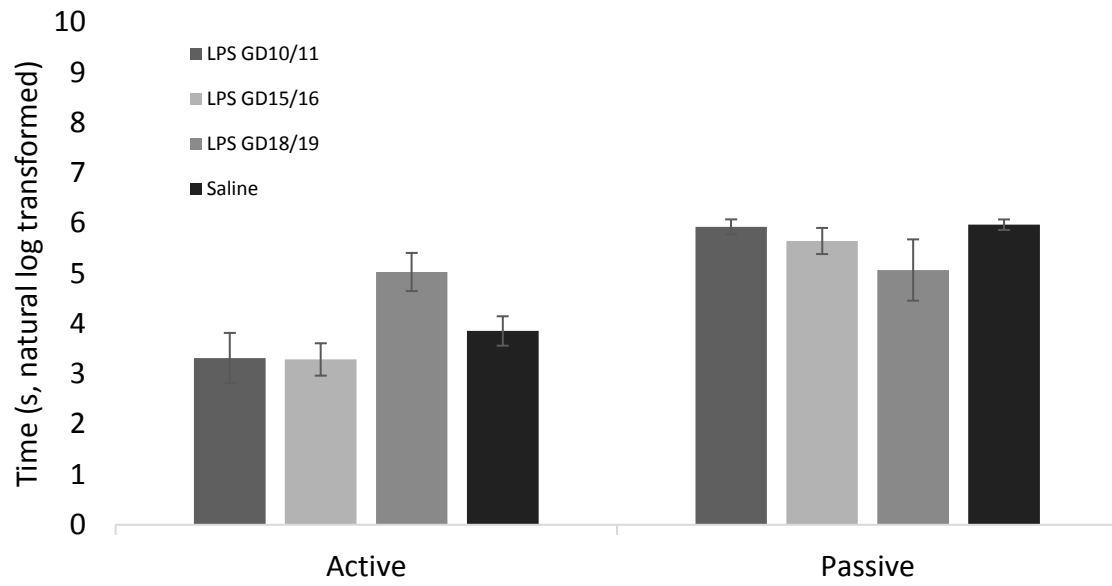


Figure 1: Duration of active and passive behaviour displayed by adolescent male rats between treatment groups. Error bars represent the standard error of the mean (SEM).

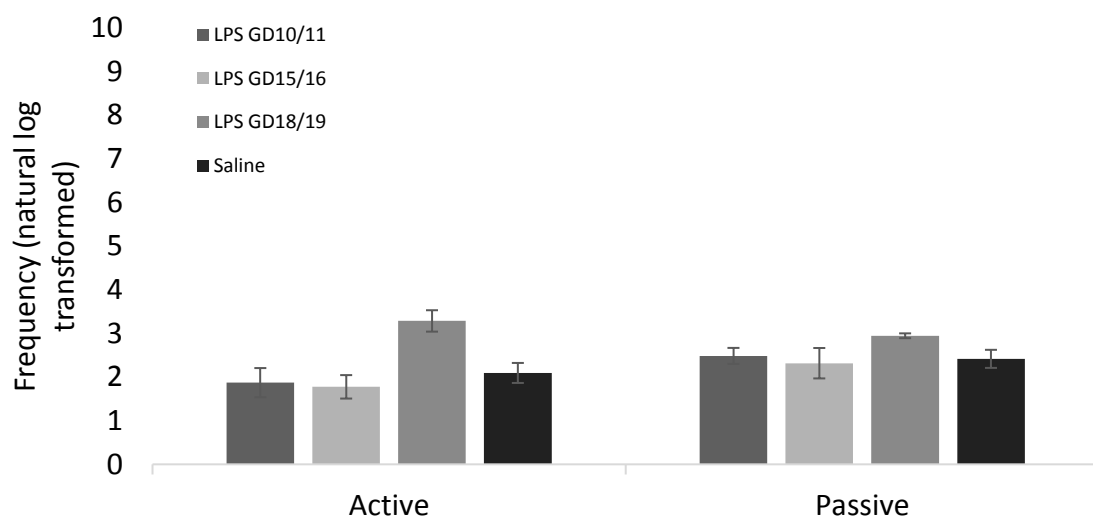


Figure 2: Frequency of active and passive behaviours displayed by adolescent male rats between treatment groups. Error bars represent SEM.

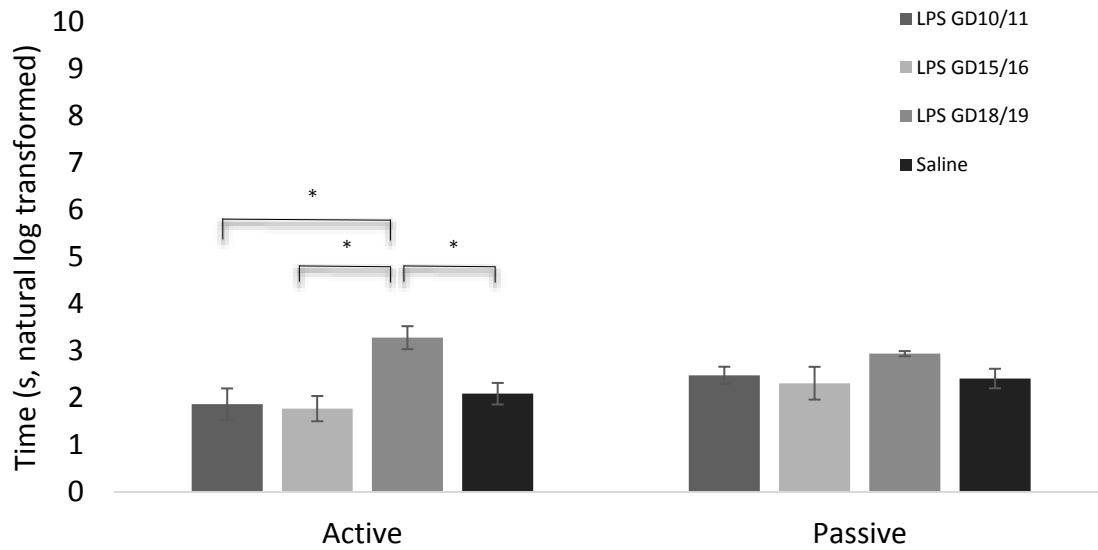


Figure 3: Latency to first active and passive behaviour displayed by adolescent male rats between treatment groups. Error bars represent SEM.

Time point 2. Figures 4, 5 and 6 show the time spent engaged in active or passive play behaviour, the frequency of these events and the latency to engage in these behaviours in the play behaviour paradigm at time point two. One-way ANOVA revealed a significant difference between conditions in the frequency of active behaviour ($F(3,35) = 5.54, p = .003$), whereby the saline condition ($M = 24.00, SD = 9.23$) interacted more frequently than LPS GD10/11 ($M = 12.13, SD = 6.29$) and LPS GD18/19 ($M = 8.00, SD = 4.76$). However, one-way ANOVA revealed no significant difference between groups in active duration ($F(3, 35) = 2.80, p = .055$), passive duration ($F(3,35) = 3.03, p = .052$), frequency of passive interaction ($F(3, 35) = .43, p = .733$), latency to first active behaviour ($F(3,35) = 1.86, p = .156$) and latency to first passive behaviour ($F(3, 35) = 1.58, p = .211$).

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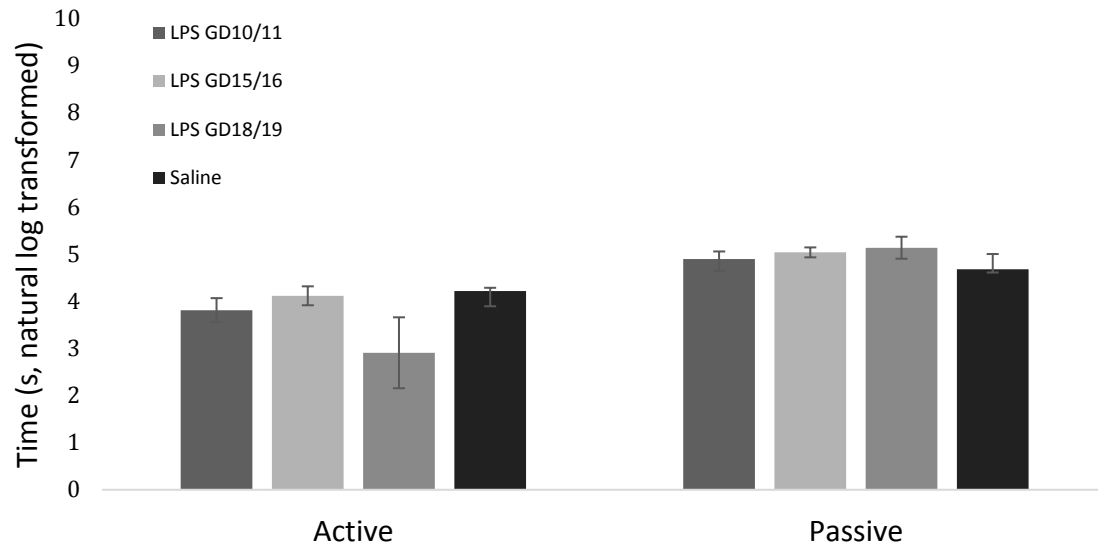


Figure 4: Duration of active and passive behaviour displayed by adult male rats between treatment groups. Error bars represent SEM.

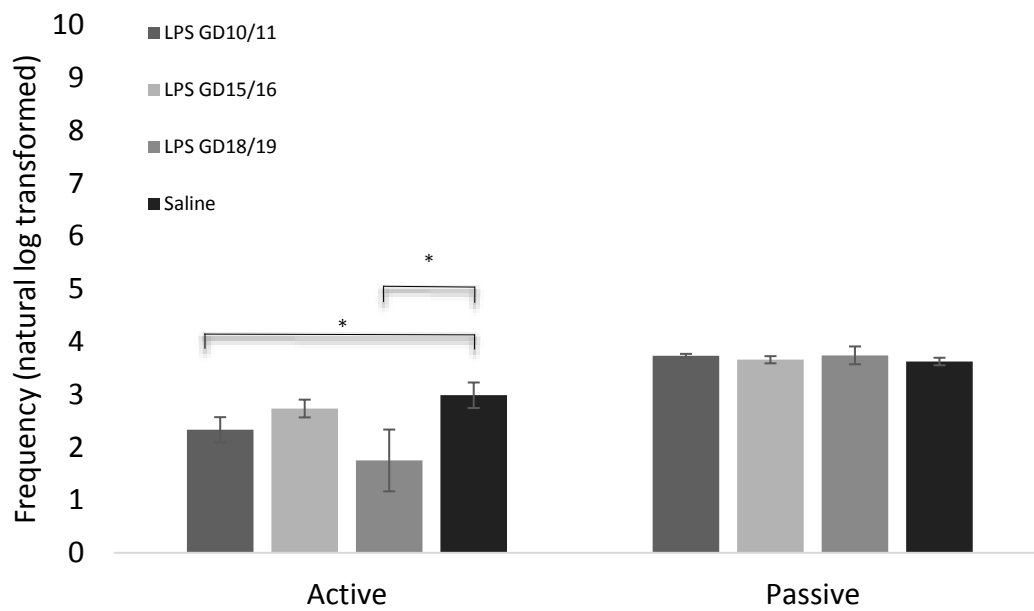


Figure 5: Frequency of active and passive behaviour displayed by adult male rats between treatment groups. Error bars represent SEM. *indicates $p < 0.05$

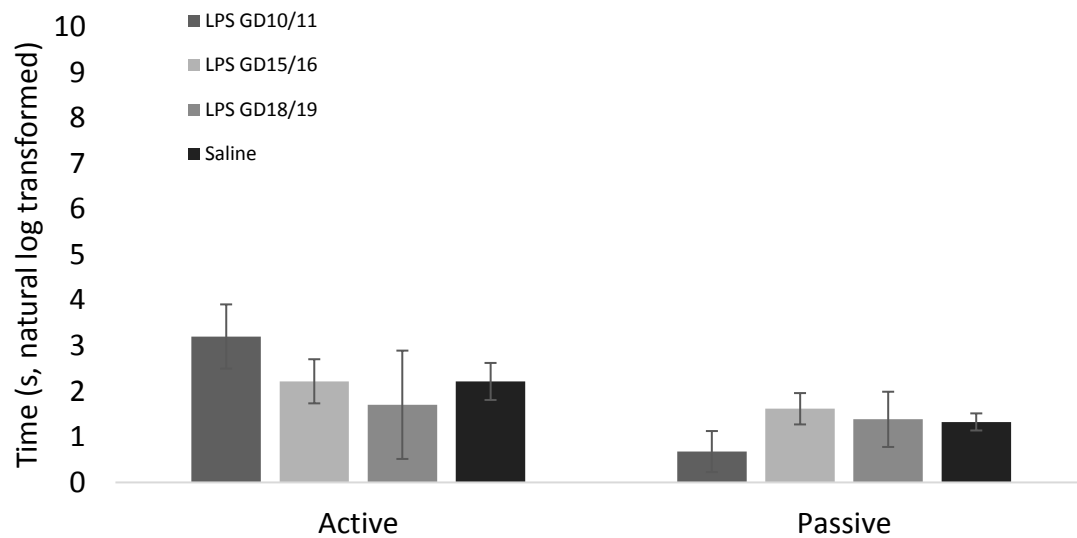


Figure 6: Latency to first active and passive behaviour displayed by adult male rats between treatment groups. Error bars represent SEM.

Social Approach and Avoidance

Time point 1. Figures 7 and 8 show the time that experimental animals spent sniffing the novel stimulus and the latency to the first sniff of the novel stimulus using the social approach and avoidance paradigm at time point two. Statistical analyses showed LPS had a significant effect on the duration spent sniffing the novel animal ($F(3, 42) = 2.84, p = .049$) (figure 7), latency to reach the novel arm ($F(3, 42) = 7.016, p = .005$) and latency to sniff the novel stimulus ($F(3, 42) = 7.02, p = .001$) (figure 8). Post hoc comparisons revealed that LPS GD15/16 rats ($M = 34.67, SD = 49.08$) spent less time sniffing the novel animal than saline controls ($M = 146.94, SD = 139.61$), took more time to reach the novel arm ($M = 5.76, SD = 1.16$) than controls ($M = 1.64, SD = 4.11$) and took more time to the first sniff of the novel stimulus ($M = 4.09, SD = 2.52$) than LPS GD10/11 ($M = 2.62, SD = 1.80$) and controls ($M = 2.97, SD = 2.36$).

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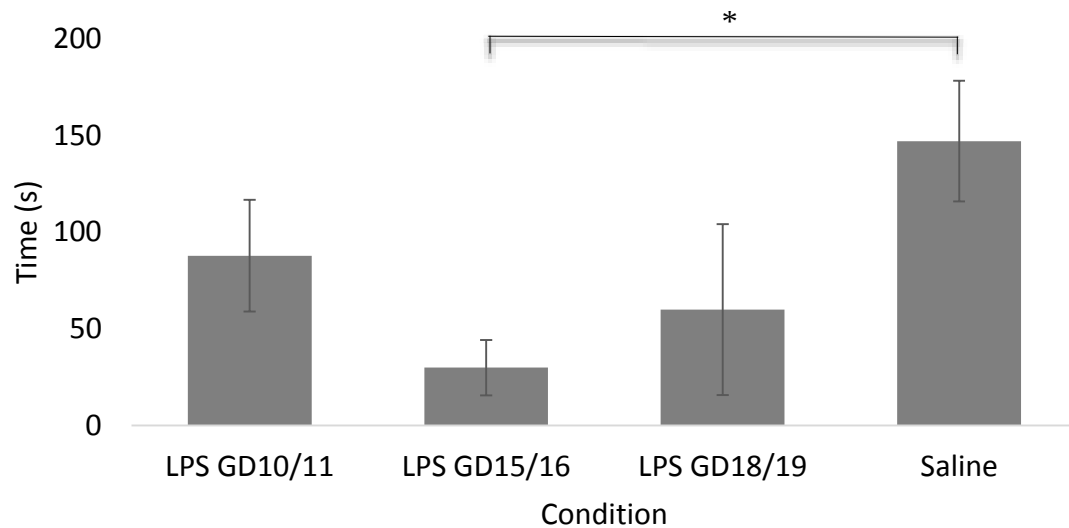


Figure 7: Duration of adolescent male rats of sniffing novel social stimulus between treatment groups. Error bars represent SEM. * indicates $p < 0.05$

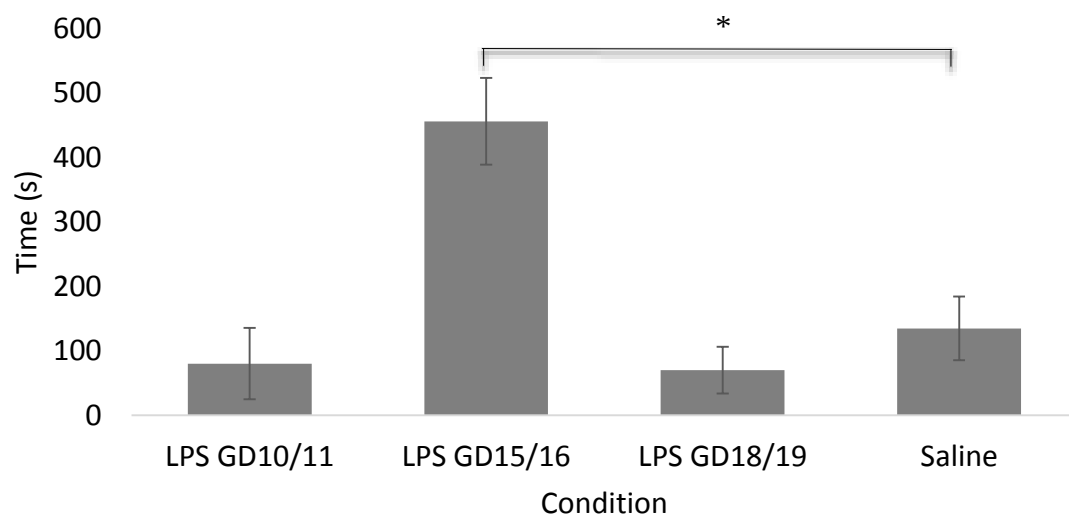


Figure 8: Latency to first sniff of novel social stimulus in adolescent male rats. between treatment groups. Error bars represent SEM. * indicates $p < 0.05$

Secondary measures showed a significant effect of LPS on distance travelled ($F(3, 42) = 4.60, p = .007$), revealed LPS GD15/16 ($M = 6.86, SD = .80$) has significantly reduced locomotor activity than saline controls ($M = 7.81, SD = .95$)

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(figure 9). LPS also had a significant effect on time spent in the non-social area, with LPS GD15/16 spending more time removed from social stimuli ($M = 4.69$, $SD = 1.91$) than saline controls ($M = 1.79$, $SD = 3.53$) (figure 10).

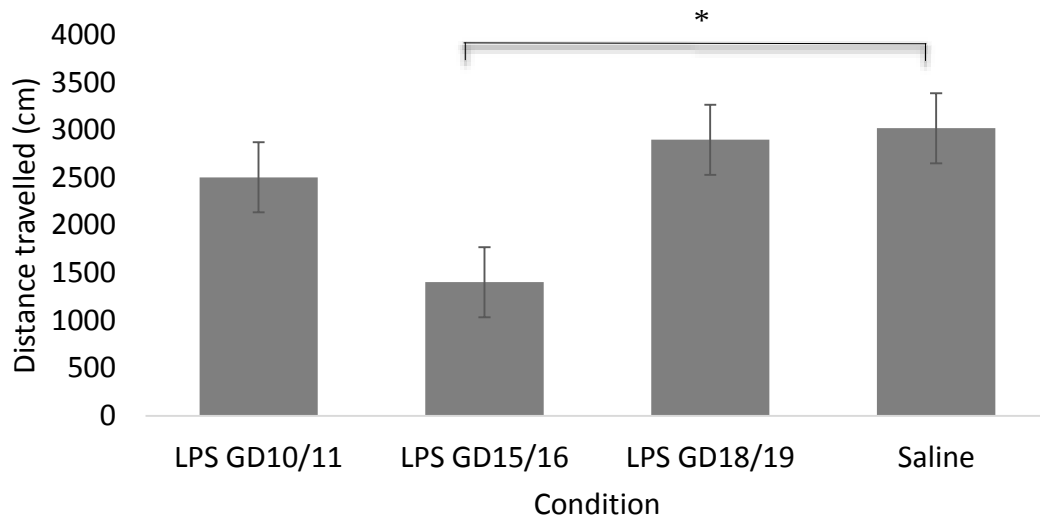


Figure 9: Mean of the total distance adolescent rat males travelled during the 10-minute test period between treatment groups. Error bars represent SEM. *indicates $p < 0.05$

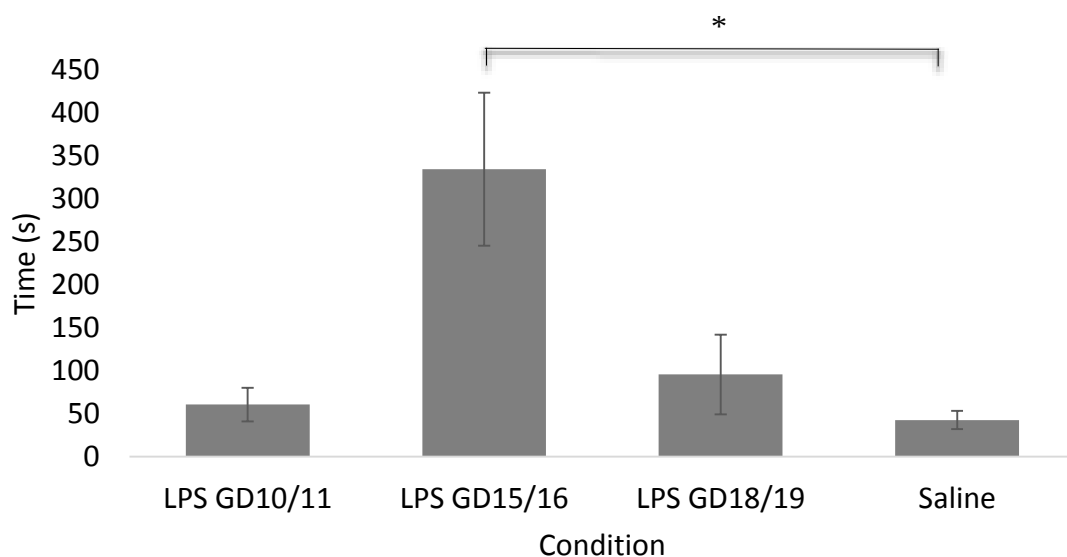


Figure 10: Duration of time that adolescent male rats spent in the non-social arm of the T-Maze between treatment groups. Error bars represent SEM. * indicates $p < 0.05$

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Time point 2. Statistical analyses revealed a significant effect of LPS on distance travelled ($F(3, 36) = 2.99, p = .044$) in the social approach avoidance paradigm at time point two (figure 11). Post hoc comparisons showed that LPS GD18/19 ($M = 8.54, SD = .82$) walked significantly further than LPS 10/11 ($M = 7.21, SD = 1.25$) and saline controls ($M = 6.87, SD = .72$). One-way ANOVAs revealed no significant differences between groups on any social measures; including duration measures, such as duration in the familiar arm ($F(3, 36) = .814, p = .316$), duration in the novel arm ($F(3, 36) = .959, p = .102$), duration sniffing the familiar stimulus ($F(3, 36) = .649, p = .553$) and duration sniffing the novel stimulus ($F(3, 36) = .932, p = .146$); including latency measures, such as latency to first entrance of familiar arm ($F(3, 36) = .237, p = .870$), latency to first entrance of novel arm ($F(3, 36) = 2.44, p = .080$), latency to first sniff of familiar stimulus ($F(3, 36) = .598, p = .621$) and latency to first sniff of the novel stimulus ($F(3, 36) = 2.99, p = .044$).

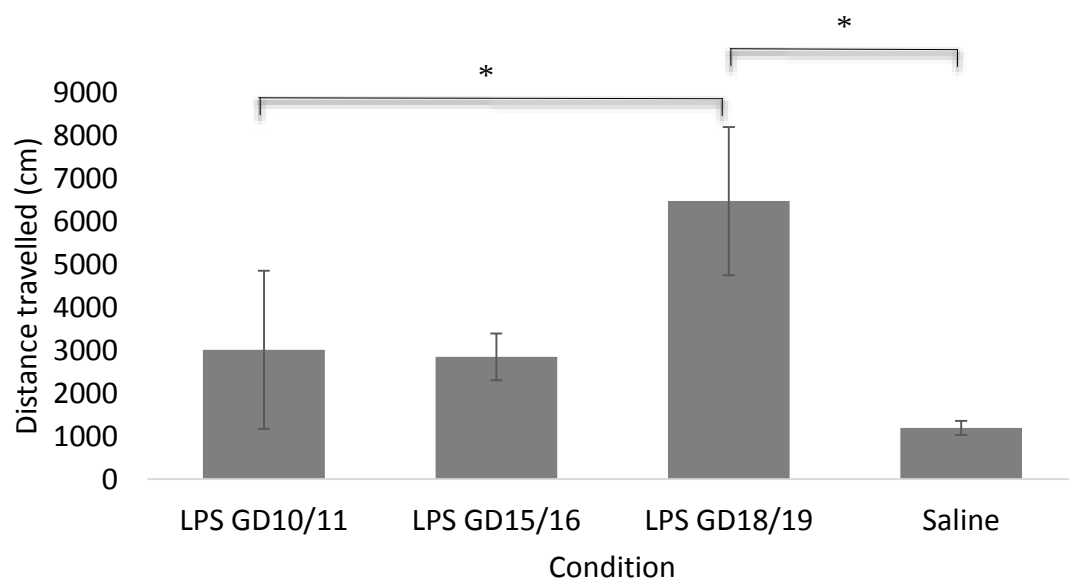


Figure 11: Mean of the total distance adult rat males travelled during the 10-minute test period between treatment groups. Error bars represent SEM.

Discussion

The development and symptomology of schizophrenia is complex. An accurate animal model of schizophrenia is required to identify environmental risk factors contributing to symptom development in humans. In turn, this will provide greater understanding of the schizophrenia phenotype and provide a valid model on which to test potential treatments and reduce the cost to both individuals and society.

Both clinical and animal research has established that prenatal maternal infection increases the risk of the offspring developing schizophrenia. Indeed, the MIA hypothesis suggests that pro-inflammatory cytokines are produced by the mother to fight the infection, but can also interfere with foetal neurodevelopment to produce neurobehavioural deficits (Gayle et al., 2004; Hava et al., 2006; Khandaker et al., 2013; Smith et al., 2007). Therefore, the MIA model is used in animal research to examine the association between prenatal infection and schizophrenia-like deficits.

The presence of cognitive and negative symptoms of schizophrenia greatly affects patient quality of life and results in poor psychosocial functioning (Ho, Nopoulos, Flaum, Arndt & Andreasen, 2004; Puig, Penadés, Gastó, Catalán, Torres & Salamero, 2008); therefore, it is integral that the development of social and cognitive deficits is greater understood. The MIA model is often used to investigate schizophrenia-like behaviours, such as acoustic startle reflex, latent inhibition and attentional set shifting; however, social functioning and social interaction are core behavioural symptoms of schizophrenia that are less well established within the MIA literature (Harvey & Boksa, 2012).

The aim of the current study was to identify whether exposure to prenatal bacterial infection during early, middle or late gestation produces deficits in social interaction. To do this we activated the maternal immune system of dams by injecting

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LPS in either early (GD10/11), mid (GD15/16) or late (GD18/19) gestation. Offspring were examined in adolescence and adulthood for deficits in social behaviour using two distinct paradigms.

Firstly, the play behaviour paradigm measures social interaction with a littermate following a period of social deprivation. The time elapsed before the first active and passive interaction, the total duration of active and passive behaviours and the frequency of active and passive behaviours were recorded (Morley-Fletcher, Rea, Maccari & Laviola, 2003). These measurements enable analysis of approach-type behaviours and social anxiety (Kirsten et al., 2010). Secondly, the social approach and avoidance paradigm measures preference for familiar and novel social stimuli (Moy et al., 2004). The T-Maze was divided into specific regions to isolate specific types of behaviour; for example, being in the vicinity of a stimulus was measured by the duration of time that the experimental animal spent in the arm holding the familiar or novel stimulus, whereas interaction required the experimental animal to sniff either stimulus through the mesh cylinder. Again, the duration of time spent in zone, frequency of entering each zone and latency to first time entering each zone was recorded. Distance travelled was also measured to provide identify alterations in locomotor activity (Yirmiya, Rosen, Donchin & Ovadia, 1994). Together, these tests aim to measure social interaction by examining active play, approach and withdrawal behaviours (Moy et al., 2004; Vanderschuren, Spruijt, Hol, Niesink & Van Ree, 1995).

LPS-exposed offspring were expected to show reduced active social behaviour in the play behaviour paradigm, specifically in GD10/11 and GD15/16 conditions. In addition, LPS-exposed offspring were expected to show avoidance from novel social stimuli, again, specifically within GD10/11 and GD15/16 conditions. These deficits in

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social behaviour are expected to be present in adolescent and adult offspring.

Consequently, it is hypothesised that prenatal MIA interferes with foetal neurodevelopment, resulting in schizophrenia-like social cognitive deficits.

Prenatal LPS and Play Behaviour

Previous literature reports that prenatal exposure to LPS in early and mid-gestation produces deficits in social behaviour in adolescent and adulthood offspring (Hava, Vered, Yael, Mordechai & Mahoud, 2006; Kirsten, Taricano, Maiorka, Palermo-Neto & Bernardi, 2010). For example, Hao et al. (2010) used LPS to initiate maternal immune activation on GD8, 9 and 12 to examine spatial learning and memory deficits; offspring developed deficits in both traits and both worsened with increased age. In contrast to this literature, the current study found that prenatal LPS-exposure resulted in no differences in active or passive social behaviours in adolescent or adult offspring.

MIA typically produces deficits in adolescence and adulthood (Kirsten et al., 2010); however, this may be due to differences found between play behaviour procedures, specifically, the length of social deprivation. Socially isolating each animal prior to testing facilitates active play behaviour; however, the duration of isolation varies from zero to 10 days (Kirsten et al., 2010). It is possible that animals exposed to LPS may respond to longer periods of social deprivation in different ways to saline exposed offspring by triggering stress responses. For example, Kirsten et al. isolated rats for 10 days prior to play behaviour testing, which may have caused an increased stress response capable of interacting with adverse effects of MIA. This interaction may trigger deficits in social behaviour, acting in accordance with the ‘two-hit’ model of schizophrenia (Maynard, Sikich, Lieberman & LaMantia, 2001).

Prenatal LPS and Social Approach and Avoidance

Deficits in social interaction are often measured using the social approach and avoidance paradigm (Moy et al., 2004). However, to our knowledge, rats prenatally exposed to LPS have not yet been assessed using this paradigm. Though, prenatal LPS-exposure has previously resulted in deficits in social interaction, locomotor activity and exploration behaviours and presentation of defensive type behaviours, which are all relevant to the social interaction of social stimuli (Elliot & Covington, 2001; Haller & Bakos, 2002; Berton, Ramos, Chaoulhoff & Mormede, 1997). GD 15/16 has been identified as a vulnerable period in regard to these behaviours (Foley et al., 2014; Lin, Lin & Wang, 2012; Oskvig et al., 2012).

In the current study two alterations were made to the typical social approach and avoidance paradigm; firstly, the period of social isolation was eliminated, instead the experimental animal entered the test phase (familiar and novel stimuli present) immediately following phase two (familiar stimulus only). Prenatal infection has previously been associated with deficits in recognition memory (Oskvig et al., 2012); therefore, this change was intended minimise the requirement for accurate recognition memory in order to isolate the experimental animal's preference for familiar or novel stimuli. Secondly, phase two of the original paradigm consists of a social stimulus held in a mesh cylinder being placed in one arm and an empty mesh cylinder placed in the other arm; therefore, allowing the experimental animal to choose between a novel social stimulus or an inanimate novel object. Instead, the mesh cylinders were present throughout all habituation and test phases; therefore, only one novel stimulus was introduced in each phase. As the mesh cylinders were present from the beginning of the procedure, they appeared as part of the T-Maze. This change places the emphasis

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on the choice between the familiar and novel social stimuli, removing the choice of social or non-social stimuli.

Therefore, considering the relevant literature and the adjusted paradigm, offspring exposed to LPS on GD15/16 were expected to show more avoidance-type behaviours, particularly from the novel social stimulus, in adolescence and adulthood.

As predicted, the current study revealed that adolescent offspring prenatally exposed to LPS on GD15/16 demonstrated avoidance-type behaviours from the novel stimulus. This behaviour was demonstrated by LPS GD15/16 spending less time in the vicinity of the novel stimulus, taking longer to enter the novel arm and taking longer to interact with the novel stimulus than the saline-treated offspring. In addition, LPS GD15/16 offspring also spent more time in the non-social arm than saline-treated offspring.

Rats are naturally socially inquisitive animals, particularly in adolescence. Therefore, normal behaviour in the social approach and avoidance paradigm includes exploring the whole T-Maze, interacting with both the familiar and novel stimuli (Moys et al., 2004; Vanderschuren, Spruijt, Hol, Niesink, & Van Ree, 1995)

In addition, LPS GD15/16 adolescent offspring also showed less locomotor activity than saline controls. Although not originally predicted, this finding may indicate reduced interest in the social stimuli, reduced exploration or increased anxiety. Oskvig et al. (2012) also found reduced locomotor activity in LPS GD15/16 offspring, suggesting that it was a socially motivated due to the experimental animals not engaging in freezing behaviours or restricted movements. Overall, LPS on GD15/16 adolescent offspring showed reduced exploration and increased social anxiety.

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It is well established in the literature that LPS-induced deficits in locomotor activity, recognition memory and spatial learning are persistent throughout adolescence and adulthood (Wischhof et al., 2014; Hao et al., 2010). However, the current findings revealed that LPS-treated adult offspring did not show any behavioural differences in their preference for familiar or novel social stimuli. Interestingly, this finding may be highlighting a methodological or treatment flaw in the current study. Alternatively, the lack of difference between LPS and saline conditions may suggest that the behavioural deficits are limited to recognition memory. Thus, by eliminating the requirement for rats to remember which animal is familiar, they are demonstrating normal social and exploratory behaviours.

Importantly, adult offspring exposed to prenatal exposure to LPS on GD18/19 showed increased locomotor activity in comparison to LPS GD10/11 and saline conditions. The expected behaviour following LPS-exposure is reduced locomotor activity, as animals are typically displaying socially anxious behaviours in the presence of an unfamiliar stimulus (Haller & Bakos, 2002; Moys et al., 2004). However, this finding has conflicted with our expectations. A potential mechanism for this behaviour is a different presentation of the same socially anxious behaviour; that is, prenatal stress can cause upregulation of the hypothalamo-pituitary-adrenal response, resulting in the offspring having long-term increased levels of stress hormones, such as corticosterone (Henry, Kabbaj, Simon, Moal & Maccari, 1994). In turn, this epigenetic change may cause increased stress and social anxiety resulting in increased locomotor activity in the novel social approach and avoidance environment (Naert, Ixart, Tapia-Arancibia & Givalois, 2006).

Overall, the current findings do reveal deficits in social interaction and interestingly, different types of deficits were evident in the three treatment groups.

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Consequently, these findings offer support that prenatal MIA interferes with foetal neurodevelopment at various stages of gestation, resulting in schizophrenia-like social deficits.

Although the current findings provided support for the hypothesis, alterations to the procedure may expose more differences between groups. First, it would be beneficial to increase the sensitivity of the behavioural measures. Increasing the period of social deprivation to 3.5 hours during play behaviour would encourage more active interaction, but should not induce an adverse environmental stress response (Kirsten et al., 2010; Vanderschuren, Spruijt, Hol, Niesink and Ree, 1996). Once animals engage in more active interaction, behaviour can be divided into specific components, such as chasing, pinning, wrestling and social grooming (File & Hyde, 1978). The current data set could not be analysed in this way due to low counts of active behaviour. However, using defined categories would provide a more sensitive measure of specific types of behaviour and results in a more detailed understanding of active play (File & Hyde, 1978). Second, to establish whether further differences exist between treatment groups, it would be beneficial to reduce the within-group variability. In the current study, animals were not handled on a regular basis; therefore, handling throughout the experiment could result in increased stress or anxiety (Ellenbroek & Yoon, 2016; Vallée et al., 1997). Handling may cause different responses between groups, but by handling animals for five minutes daily leading up to each experiment, any differential stress effects could be minimised. In addition, testing each animal on three consecutive days and using an average score would minimise variability due to changes in uncontrollable variables, such as noise from outside the experimental room.

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Despite the large within-group variability, the current findings have demonstrated that early and mid-gestation are a high-risk period of development. However, it remains unclear what mechanism contributes to these effects. The neurodevelopmental hypothesis suggests that challenge, such as MIA, during various critical phases of brain development, such as myelination and synaptic pruning, can result in long-term changes in behaviour (Wischhof et al., 2015; Zuckerman et al., 2003). Whereas, a leading hypothesis of autism posits that offspring are most vulnerable to valproate during neural tube closure, occurring at GD12.5 in rats and between weeks three and four in humans (Kim et al., 2011). Schizophrenia and neural tube deficits has been reviewed; however, periods of increased vulnerability towards LPS have not yet been explored in relation to neural tube closure.

Investigating specific periods of vulnerability is an important feature of animal research. However, a large variety of methods are currently used to induce prenatal infection making it difficult to compare the effects of MIA and LPS across studies. Firstly, there is variation in the method of administration; intraneuronal (Zhu, Zhang, Ding, Zhao & Zhang, 2014), intraperitoneal (Oskvig et al., 2012) and subcutaneous (Waterhouse et al., 2016) LPS-injections have all significantly altered foetal neurodevelopment and offspring behaviour. Secondly, doses range from 100µg to 500µg and lastly, frequency of injecting LPS varies from one (Taylor et al., 2012) to every day during pregnancy (for review Boksa, 2010). Variation in the method of infection, the frequency and the severity could lead to different neurochemical and behavioural deficits between offspring conditions. More data is required to establish the specific the impact of these differences in methodology; however, this must still be considered when analysing MIA and schizophrenia literature.

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To develop the animal model of schizophrenia, research must not only focus upon developing an accurate and reliable model of MIA, but ensure that it is translatable to clinical research and practise (Van der Worp et al., 2010). As previously acknowledged, the emphasis on schizophrenia development in humans remains on genetic and environmental interactions and the ‘two-hit model’; however, there is no genetic manipulation and only one exposure to environmental challenge used within the current study (Maynard et al., 2001; Walder et al., 2014). Examining prenatal MIA in conjunction with an additional risk factor may increase the severity of deficits. For example, due to schizophrenia being associated with the dysfunction of the dopaminergic system (Zuckerman et al., 2003), future research could investigate the vulnerability D1 mutants have to prenatal LPS-exposure. In addition, manipulating postnatal environment by inducing stress (Haller & Bakos, 2002) or infection (Zhu et al., 2014) in the offspring may also increase such deficits. This may be particularly interesting during adolescence, due to the increased vulnerability to environmental stressors (Walder et al., 2014).

Schizophrenia is characterised by onset between late adolescence to early adulthood; however, negative symptoms, such as reduced social behaviour can be reported prior to the onset of psychotic symptoms (Lencz, Smith, Auther, Correll, & Cornblatt, 2004; Puig et al., 2008). As a result, it was important that social behaviour was measured at two distinct time points during the current study. However, schizophrenia has also been associated with delayed achievement of developmental milestones (Sorensen et al., 2010). Therefore, to develop a valid and holistic animal model of schizophrenia it would be beneficial to include measurements of physiological development, such as changes in body weight, the opening of eyes, fingers and ears and additional behavioural measurements, such as food and water

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intake (Asiaei, Solati & Salari, 2011). To provide further support the MIA model of schizophrenia, early prenatal LPS-exposure would be expected to associate with delays in developmental milestone (Asiaei et al., 2011; Sorensen et al., 2010).

Overall, the current study did identify social behaviour deficits, particularly in LPD GD15/16 condition, suggesting that LPS-exposure did cause MIA, increased cytokine production and disruption in neurodevelopment. However, no physiological or neurochemical measurements were taken to ensure activation of the maternal immune system. A manipulation check could be done using various methods. Firstly, maternal physiological or immune response can be measured using temperature, heart rate or neural cytokine increase following exposure to LPS and saline, suggesting fever was present in the dams (Gayle, 2004). Second, offspring can be sacrificed and brains examined for increases in foetal brain cytokines to ensure neurochemical changes have occurred (Zhu et al., 2014). However, Kirsten et al. (2010) identified behavioural but no morphological differences in LPS-exposed offspring on PND60. Last, it is well established in the literature that prenatal LPS-exposure produces deficits in prepulse inhibition; therefore, replicating this behavioural test would provide a manipulation check, confirming the adverse effect of prenatal LPS.

Overall, this study shows that MIA can induce schizophrenia-like deficits, specifically when exposure to LPS occurred in early to mid gestation. This demonstrates that prenatal environment is a sensitive period of neurodevelopment and immune challenge during this period may contribute to the social deficits present in schizophrenia.

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